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VU Answer

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Rephrase the following statement in bi-conditional form
"If you get up early in the morning, you will be healthy"

Answer (Please select your correct option)

☐ You will be healthy if and only if you get up early in the morning

correct

☐ If you will be healthy then you will get up early in the morning

☐ If you will get up early in the morning then you will be healthy

☐ None of these

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Reductio ad absurdum law is symbolically denoted as

Answer (Please select your correct option)

☐

$$(p \wedge q) \rightarrow r \equiv p \rightarrow (q \rightarrow r)$$

☐

$$p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$$

☐

$$p \rightarrow q \equiv \sim p \vee q$$

☐

$$p \rightarrow q \equiv (p \wedge \sim q) \rightarrow c$$

correct

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A critical row is that in which _____ premises have truth value T.

Answer (Please select your correct option)

☐ at least one

☐ exactly one

☒ all

correct

☐ at least two



What will be the output of an OR-gate if it has inputs 0 and 1?

Answer (Please select your correct option)

☐ 0

☒ 1

☐ 2

☐ 3

correct



Let U be the universal set and A is its subset then $A \cup A^c$ is equal to

Answer (Please select your correct option)

☐ A

correct

☐ A^c ☐ ϕ ☐ U

Identify the false statement

Answer (Please select your correct option)

☐

$0 \in \phi$

correct

☐

$\{\phi\} \subseteq \{\phi\}$

☐

If A and B are two sets $A \subseteq B$ and $B \subseteq A$ then $A = B$

☐

Two sets are disjoint if their intersection is empty set

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If A and S are two reflexive relations then $A \cap S$ will be

Answer (Please select your correct option)

☐ Symmetric

☐ Reflexive

correct

☐ Transitive



Symmetric and Anti-symmetric relations are

Answer (Please select your correct option)

☐

negative of each other.

☐

same.

☐

not negative of each other.

correct

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If two relations are reflexive then their composition is

Answer (Please select your correct option)

☐

Anti-symmetric

☐

Reflexive

correct

☐

Irreflexive

☐

Symmetric

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Inverse of relation can be obtained by

Answer (Please select your correct option)

☐

changing signs of elements in order pairs.

☐

changing position of elements in order pairs.

correct

☐

taking multiplicative inverse of elements in order pairs.

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Let $A \times A = \{(1,1), (1,2), (1,3), (2,1), (2,2), (2,3), (3,1), (3,2), (3,3)\}$, find which one of the given relations is a function.

Answer (Please select your correct option)

☐

$$R_1 = \{(1,3), (2,2), (3,1)\}$$

☐

$$R_2 = \{(1,1), (1,2), (2,1)\}$$

☐

$$R_3 = \{(2,2), (2,3), (3,1)\}$$



If $f(x) = 2x + 1$ and $g(x) = x^2 - 1$ then $f \circ f(x) =$

Answer (Please select your correct option)



$4x - 3$



$4x^2 + 1$



$4x + 3$

correct



$4x^2 - 1$



Let f and g be the functions defined by $f(x) = 2x + 3$ and $g(x) = 3x + 2$ then composition of f and g is

Answer (Please select your correct option)

☐ $6x + 6$

correct

☐ $5x + 5$

☐ $6x + 7$



The negation of $1 < x < 10$ is $x \leq 1$ or $x \geq 10$ by using:

Answer (Please select your correct option)

☐ Distributive Law

☐ Inequality Law

☐ De-Morgan's Law

☐ None of these



If the n th term of a sequence is $a_n = 2(-3)^n + 5^n$ then the term a_1 is

Answer (Please select your correct option)

☐ -1

☐ 0

☐ 1

☐ 2



The part of definition which can be expressed in terms of smaller versions of itself is called

Answer (Please select your correct option)

☐ Recursion

correct

☐ Conclusion

☐ Base

☐ Restriction

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The tower of Hanoi is a puzzle consisting of

Answer (Please select your correct option)



2 people



3 people

correct



4 people



The same element can never appear ----- in a set.

Answer (Please select your correct option)

twice



correct

once



thrice



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If $(A \cup B) = A$ then

Answer (Please select your correct option)

☐

$$(A \cap B) = B^c$$

☐

$$(A \cap B) = A$$

☐

$$(A \cap B) = B$$

correct

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If p = It is raining

q = She will go to college

"It is raining and she will not go to college" will be denoted by

Answer (Please select your correct option)

☐ $p \wedge \neg q$

correct

☐ $p \wedge q$

☐ $\neg (p \wedge q)$

☐ $\neg p \wedge q$